

- All problems are worth an equal weight, even if some may be harder than others.
- Write answers on a separate sheet, either typed or handwritten, either on paper or a note taking device. No need to copy the problem statements. Please, upload an unique pdf file.
- Late submissions will be worth significantly less points.
- Circle your final answer, but make sure to include the work to get there.
- It is OK to work with others, as long as you acknowledge them in your paper and write down your solutions in your own words. It is not OK to copy a solution from whatever source.

For problems 1-7, find the eigenvalues λ and a complete set of eigenvectors of the given matrix A. For each eigenvector v found, verify that $Av = \lambda v$. Eigenvalues may be complex numbers, in which case your eigenvectors will also involve complex numbers.

1. $\begin{bmatrix} 6 & -3 \\ 2 & -1 \end{bmatrix}$

2. $\begin{bmatrix} 5 & 4 \\ -10 & -7 \end{bmatrix}$

3. $\begin{bmatrix} 0 & -2 \\ 8 & 0 \end{bmatrix}$

4. $\begin{bmatrix} -11 & 9 \\ -16 & 13 \end{bmatrix}$

5. $\begin{bmatrix} -8 & 6 & -2 \\ -6 & 4 & -2 \\ 9 & -6 & 3 \end{bmatrix}$

6. $\begin{bmatrix} -2 & 1 & -1 \\ -2 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}$

7. $\begin{bmatrix} 3 & 0 & 1 \\ 0 & -4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

For problems 8-10, find a 2×2 matrix that satisfies the conditions given.

In order to do so, call its elements x, y, z, w and obtain equations to solve for these unknowns from $Av = \lambda v$.

8. Two real eigenvalues 2 and -1 , with corresponding eigenvectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$.

9. Two complex eigenvalues $2 + i$ and $2 - i$, with corresponding eigenvectors $\begin{bmatrix} 1 + 2i \\ 2 + 3i \end{bmatrix}$ and $\begin{bmatrix} 1 - 2i \\ 2 - 3i \end{bmatrix}$.

10. Only one repeated eigenvalue 3, with 1-dimensional eigenspace spanned by the eigenvector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.